L-TORSION OF THE PICARD GROUP OF A REDUCED CONNECTED PROPER CURVE IS BOUNDED BY GENUS PLUS GEOMETRIC GENUS, WITH EQUALITY EXACTLY FOR MULTICROSS SINGULARITIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the three-step factorization with its Picard sequences and cohomology changes, the l-divisibility computation, the endpoint values on the normalization, the counting bound, and the two directions of the equality characterization are retained verbatim, including the author's remark that the final gluing of local sequences is given as a hint with some details omitted.

Editorial note. The label proposition-torsion-picard-reduced-proper names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own proposition environment in the chapter source.

Allowed standard prerequisites (editorial).

- Factorization of the normalization into elementary modifications. Same chapter, Lemma lemma-factor-almost-isomorphism with Lemmas lemma-glue-points and lemma-squish-tangent-vector, and Examples example-glue-points, example-squish-tangent-vector. Supplies the sequence X_n -> ... -> X_0 = X where each step glues two points or squishes a tangent vector, with the induced exact sequences on Picard and the changes of h^0, h^1.
- Torsion Picard of a smooth projective curve. Same chapter, Lemma lemma-torsion-picard-smooth-projective; Varieties, Lemma varieties-lemma-prepare-delta-invariant. Gives the endpoint values on the normalization X_n = disjoint union of smooth projective curves: h^1_n = g_geom and dim Pic(X_n)[l] = 2 g_geom.

Source reference. Original reference 0C1J: sources/stacks-curves/pinned-original-source.tex, source lines 3202–3202. Prerequisite pointer invoked at proof line(s) 3749; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C1K: sources/stacks-curves/pinned-original-source.tex, source lines 3228–3228. Prerequisite pointer invoked at proof line(s) 3749; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original reference 0C1L: sources/stacks-curves/pinned-original-source.tex, source lines 3257–3257. Prerequisite pointer invoked at proof line(s) 3722; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C1M: sources/stacks-curves/pinned-original-source.tex, source lines 3323–3323. Prerequisite pointer invoked at proof line(s) 3725; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C1X: sources/stacks-curves/pinned-original-source.tex, source lines 3478–3478. Prerequisite pointer invoked at proof line(s) 3776; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C1N: sources/stacks-curves/pinned-original-source.tex, source lines 3364–3364. Prerequisite pointer invoked at proof line(s) 3725; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C1Z: sources/stacks-curves/pinned-original-source.tex, source lines 3611–3611. Prerequisite pointer invoked at proof line(s) 3766; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C1R: sources/stacks-chow/tags.txt, source lines 15131–15131. Prerequisite pointer invoked at proof line(s) 3717; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *Let k be an algebraically closed field. Let X be a proper scheme over k which is reduced, connected, and has dimension 1. Let g be the genus of X and let g_{geom} be the sum of the geometric genera of the irreducible components of X. For any prime ℓ different from the characteristic of k we have*

$$\dim_{\mathbf{F}_\ell} \mathrm{Pic}(X)[\ell] \leq g + g_{geom}$$

and equality holds if and only if all the singularities of X are multicross.

Proof. Let $\nu : X^\nu \to X$ be the normalization (Varieties, Lemma 0C1R). Choose a factorization

$$X^\nu = X_n \to X_{n-1} \to \ldots \to X_1 \to X_0 = X$$

as in Lemma 0C1L. Let us denote $h_i^0 = \dim_k H^0(X_i, \mathcal{O}_{X_i})$ and $h_i^1 = \dim_k H^1(X_i, \mathcal{O}_{X_i})$. By Lemmas 0C1M and 0C1N for each $n > i \geq 0$ we have one of the following there possibilities

(1) X_i is obtained by glueing $a, b \in X_{i+1}$ which are on different connected components: in this case $\mathrm{Pic}(X_i) = \mathrm{Pic}(X_{i+1})$, $h_{i+1}^0 = h_i^0 + 1$, $h_{i+1}^1 = h_i^1$,
(2) X_i is obtained by glueing $a, b \in X_{i+1}$ which are on the same connected component: in this case there is a short exact sequence

$$0 \to k^* \to \mathrm{Pic}(X_i) \to \mathrm{Pic}(X_{i+1}) \to 0,$$

and $h_{i+1}^0 = h_i^0$, $h_{i+1}^1 = h_i^1 - 1$,

(3) X_i is obtained by squishing a tangent vector in X_{i+1}: in this case there is a short exact sequence

$$0 \to (k, +) \to \mathrm{Pic}(X_i) \to \mathrm{Pic}(X_{i+1}) \to 0,$$

and $h^0_{i+1} = h^0_i$, $h^1_{i+1} = h^1_i - 1$.

To prove the statements on dimensions of cohomology groups of the structure sheaf, use the exact sequences in Examples 0C1J and 0C1K. Since k is algebraically closed of characteristic prime to ℓ we see that $(k, +)$ and k^* are ℓ-divisible and with ℓ-torsion $(k, +)[\ell] = 0$ and $k^*[\ell] \cong \mathbf{F}_\ell$. Hence

$$\dim_{\mathbf{F}_\ell} \mathrm{Pic}(X_{i+1})[\ell] - \dim_{\mathbf{F}_\ell} \mathrm{Pic}(X_i)[\ell]$$

is zero, except in case (2) where it is equal to -1. At the end of this process we get the normalization $X^\nu = X_n$ which is a disjoint union of smooth projective curves over k. Hence we have

(1) $h^1_n = g_{geom}$ and
(2) $\dim_{\mathbf{F}_\ell} \mathrm{Pic}(X_n)[\ell] = 2g_{geom}$.

The last equality by Lemma 0C1Z. Since $g = h^1_0$ we see that the number of steps of type (2) and (3) is at most $h^1_0 - h^1_n = g - g_{geom}$. By our comptation of the differences in ranks we conclude that

$$\dim_{\mathbf{F}_\ell} \mathrm{Pic}(X)[\ell] \leq g - g_{geom} + 2g_{geom} = g + g_{geom}$$

and equality holds if and only if no steps of type (3) occur. This indeed means that all singularities of X are multicross by Lemma 0C1X. Conversely, if all the singularities are multicross, then Lemma 0C1X guarantees that we can find a sequence $X^\nu = X_n \to \ldots \to X_0 = X$ as above such that no steps of type (3) occur in the sequence and we find equality holds in the lemma (just glue the local sequences for each point to find one that works for all singular points of x; some details omitted). □